



VPRO SKILLS

FAQ / INTERVIEW PREPARATION

Vectorless Interview Questions & Answers

STUDENT LEARNING RESOURCE

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How to Use This Guide

This handbook brings together 24 curated questions and answers on Vectorless, organized into a single quick-reference list for focused revision. Each question is numbered for easy reference, and answers are broken down into definitions, examples, code, and comparison tables wherever the source material provides them.

- Code blocks are shown in a monospaced, shaded panel — read them alongside the explanation above.
- Comparison tables summarize key differences at a glance; use them for quick revision before an interview.
- The Quick Revision page at the end lists every question so you can self-test your recall.

VECTORLESS

Q01. What is Vectorless RAG?

Vectorless RAG is a RAG approach where we retrieve information without converting documents into embeddings and without using a vector database.

It can use methods such as:

- Keyword search
- Exact text matching
- Full-text search
- BM25

Simple flow:

```
Document
  ↓
Text chunks
  ↓
Text Search
  ↓
Relevant chunks
  ↓
LLM
  ↓
Answer
```

Q02. Why would you use Vectorless RAG instead of Vector RAG?

I would use Vectorless RAG when the data can be retrieved effectively using text-based search.

For example:

```
Question:
"What is the employee leave policy?"
Document:
"Employees are allowed 20 days of leave per year."
```

Searching for words like:

```
employee
leave
policy
```

can directly find the relevant document.

So we don't necessarily need embeddings and a vector database.

Q03. Why not always use Vector RAG?

Because embeddings and vector databases add extra components.

Vector RAG requires:

```
Embedding Model
      ↓
Vector Database
      ↓
Similarity Search
```

Vectorless RAG can be:

```
Text
  ↓
Search
  ↓
Relevant Text
```

If simple text search already gives good results, adding vectors may add unnecessary complexity.

Q04. Does Vectorless RAG mean there is no search?

No.

There is still retrieval.

The difference is how we retrieve the information.

```
Vector RAG
→ Similarity / semantic search
Vectorless RAG
→ Keyword / text / full-text search
```

Q05. What is the biggest limitation of Vectorless RAG?

The biggest limitation is that simple keyword search may not understand the meaning of a question.

EXAMPLE

```
Document:
"The employee can work remotely."
Question:
"Can I work from home?"
```

The document says remotely, while the question says work from home.

A simple keyword search may not find it.

Vector search can potentially find it because the meanings are similar.

Q06. Which has better performance: Vector RAG or Vectorless RAG?

There is no universal winner.

Performance depends on:

- Dataset size
- Search engine
- Query type
- Number of documents
- Search algorithm
- Infrastructure

For simple text matching, Vectorless RAG can be fast and lightweight.

For semantic questions, vector search can provide better retrieval because it searches based on similarity of meaning.

Q07. Is Vectorless RAG faster?

It can be, especially for smaller datasets and simple searches.

Vectorless RAG does not need:

```
Embedding generation
Vector storage
Vector similarity calculation
```

But performance depends on the search technology.

For large datasets, a properly indexed full-text search system can also perform very efficiently.

Q08. Is Vectorless RAG cheaper?

It can reduce costs because we may not need:

- An embedding model
- Embedding generation
- Vector database
- Vector storage

But the actual cost depends on the search infrastructure and scale.

So I would say:

Vectorless RAG can be cheaper when the application does not need semantic vector search.

Q09. What happens during indexing in Vectorless RAG?

A simple implementation might do:

```
Documents
  ↓
Clean text
  ↓
Split into chunks
  ↓
Store text
  ↓
Create text-search index
```

There is no embedding generation.

Q10. What happens during retrieval?

The user asks a question.

For example:

```
"What is the refund policy?"
```

The system searches the indexed text for relevant terms:

```
refund
policy
```

Then it returns the most relevant chunks.

```
Question
  ↓
Text Search
  ↓
Top Relevant Chunks
  ↓
LLM
```

Q11. What search algorithms can be used in Vectorless RAG?

Common approaches include:

```
Keyword Search
Exact Match
Full-Text Search
BM25
```

For production systems, BM25 and full-text search are common choices when lexical retrieval is appropriate.

Q12. What is BM25?

BM25 is a text-ranking algorithm.

It helps answer:

Q13. Which documents are most relevant to these search words?

For example:

Query:
Python functions

BM25 can rank documents containing those terms based on factors such as:

- How often the words occur
- How important the words are
- Document length

The result is a ranked list of documents.

Q14. Vector RAG vs Vectorless RAG — which is better for semantic search?

Vector RAG is designed for semantic similarity.

Vectorless keyword-based retrieval is primarily based on text matching.

EXAMPLE

Question:
"How can I work from home?"
Document:
"Employees can work remotely."

Semantic retrieval can recognize that these concepts are related.

Simple keyword search may not.

Q15. When is Vectorless RAG a good choice?

Good use cases include:

FAQ systems

"What is the refund policy?"

Documentation search

"How do I reset my password?"

Legal/document search

"termination clause"

Company policies

"leave policy"

Product documentation

"database connection configuration"

When users normally use the same terminology as the documents, text search can work well.

Q16. When should you prefer Vector RAG?

When users ask questions using different words from the documents.

EXAMPLE

Document:
"Employees may work remotely."
Question:
"Can I work from home?"

Here, semantic retrieval can be useful because the wording is different but the meaning is related.

Q17. Can Vectorless RAG handle large datasets?

Yes.

But we should not simply scan every document manually.

For large datasets, we should use an efficient search engine and indexes.

For example:

Millions of documents
↓
Search Index
↓
BM25 / Full-text Search
↓
Top relevant documents

So:

Vectorless does not mean "search every document one by one."

Q18. What is the performance bottleneck in Vectorless RAG?

It depends on the implementation.

A poor implementation might do:

```
for document in all_documents:  
    search(document)
```

This becomes inefficient as data grows.

A production implementation should use an indexed search engine.

```
Documents  
  ↓  
Search Index  
  ↓  
Fast Retrieval
```

Q19. What is the main trade-off between Vector and Vectorless RAG?

The trade-off is mainly:

```
Vector RAG  
→ Better semantic understanding  
→ More infrastructure  
Vectorless RAG  
→ Simpler retrieval  
→ More dependent on matching the document's terminology
```

Q20. Can Vectorless RAG completely replace Vector RAG?

No.

They solve different retrieval problems.

If the application mainly needs:

```
Exact terms  
Known terminology  
Keyword search  
Document lookup
```

Vectorless RAG can be sufficient.

If the application needs:

Meaning-based search
Different wording
Semantic similarity

Vector RAG may be more suitable.

Q21. What would you choose for an enterprise RAG application?

I would not decide based only on whether vectors are popular.

I would first test the retrieval requirements.

I would evaluate:

1. What type of questions do users ask?
2. How large is the data?
3. Do users use the same terminology as documents?
4. Do we need semantic search?
5. What retrieval accuracy do we need?
6. What is the latency requirement?
7. What is the infrastructure cost?

Then I would select the retrieval approach based on the results.

Q22. Experienced interviewer: "Why did you choose Vectorless RAG?"

Strong answer:

"I chose Vectorless RAG because the application primarily required lexical retrieval rather than semantic similarity. The documents had predictable terminology, so BM25/full-text search could retrieve the required content without introducing an embedding model and vector database. This simplified the architecture and reduced the indexing pipeline. I would consider vector or hybrid retrieval if query wording differed significantly from the source documents."

Q23. Experienced interviewer: "How would you measure its performance?"

I would measure both retrieval quality and system performance.

Retrieval quality

Precision
Recall
MRR

System performance

Latency
Throughput
Indexing time
Storage
Cost

For example:

Query
↓
Retrieve Top-K documents
↓
Check whether the correct document was retrieved
↓
Measure retrieval quality

Q24. Experienced interviewer: "What if keyword search is not giving good results?"

I would not immediately replace the entire system.

I would investigate:

Query processing
↓
Text normalization
↓
Search configuration
↓
BM25 tuning
↓
Chunking
↓
Top-K selection

If semantic matching is still required, I could introduce:

Hybrid Retrieval
Keyword Search
+
Vector Search
↓
Combined Results

One Interview Formula to Remember

```
VECTOR RAG
Search by MEANING
  ↓
Embeddings
  ↓
Vector Database
  ↓
Similarity Search
VECTORLESS RAG
Search by TEXT
  ↓
Keyword / BM25
  ↓
Text Search
  ↓
Relevant Chunks
```

The most important interview question:

"Why Vectorless RAG?"

Remember this answer:

"If my application can retrieve the required information effectively through text-based search, I don't need to introduce embeddings and a vector database. Vectorless RAG gives me a simpler retrieval architecture. If semantic matching becomes important, I can consider vector or hybrid retrieval."

Quick Revision

Scan through these questions to self-test your recall before an interview.

- Q01. What is Vectorless RAG?
- Q02. Why would you use Vectorless RAG instead of Vector RAG?
- Q03. Why not always use Vector RAG?
- Q04. Does Vectorless RAG mean there is no search?
- Q05. What is the biggest limitation of Vectorless RAG?
- Q06. Which has better performance: Vector RAG or Vectorless RAG?
- Q07. Is Vectorless RAG faster?
- Q08. Is Vectorless RAG cheaper?
- Q09. What happens during indexing in Vectorless RAG?
- Q10. What happens during retrieval?
- Q11. What search algorithms can be used in Vectorless RAG?
- Q12. What is BM25?
- Q13. Which documents are most relevant to these search words?
- Q14. Vector RAG vs Vectorless RAG — which is better for semantic search?
- Q15. When is Vectorless RAG a good choice?
- Q16. When should you prefer Vector RAG?
- Q17. Can Vectorless RAG handle large datasets?
- Q18. What is the performance bottleneck in Vectorless RAG?
- Q19. What is the main trade-off between Vector and Vectorless RAG?
- Q20. Can Vectorless RAG completely replace Vector RAG?
- Q21. What would you choose for an enterprise RAG application?
- Q22. Experienced interviewer: "Why did you choose Vectorless RAG?"
- Q23. Experienced interviewer: "How would you measure its performance?"
- Q24. Experienced interviewer: "What if keyword search is not giving good results?"